MA416 Project 1 – MLB Pitching Performance Analysis

AL Central Teams and Time Through the Order, 2024 Season

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1 Introduction

This report analyzes MLB Statcast pitch-by-pitch data from the 2024 season, aggregated to the at-bat level. The dataset contains one row per at-bat, with the last pitch of each at-bat carrying the outcome. The primary response variable is `estimated_woba_using_speedangle` (xwOBA on contact), a Statcast metric that measures the expected weighted on-base average based on exit velocity and launch angle on balls put in play. Missing xwOBA values correspond to at-bats without contact (strikeouts, walks, HBP) and are excluded from analysis.

Two inferential questions are investigated:

1. **Section 1:** Do AL Central pitching staffs differ in the quality of contact they allow? A one-factor ANOVA compares mean xwOBA across five AL Central teams (CLE, CWS, DET, KC, MIN).
2. **Section 2:** Does a pitcher's performance degrade as batters face them additional times through the batting order? A repeated-measures ANOVA compares mean xwOBA allowed by individual pitchers across the first, second, and third time through the order (TTO 1, 2, 3).

In both cases, assumption violations motivate a non-parametric permutation test as a distribution-free follow-up (Section 3).

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2 Section 1: One-Factor ANOVA — AL Central Pitching

2.1 Design and Methodology

Groups (g = 5): Kansas City Royals (KC), Detroit Tigers (DET), Minnesota Twins (MIN), Chicago White Sox (CWS), Cleveland Guardians (CLE).

At-bat assignment: Each at-bat is assigned to the pitcher's team. When the inning is the top half, the home team is pitching; otherwise, the away team is pitching. Only at-bats where the pitcher's team is one of the five AL Central teams are included.

Multi-team pitchers: Any pitcher who appeared for more than one AL Central team in 2024 is excluded to maintain clean group membership.

Response: `estimated_woba_using_speedangle` (xwOBA on contact). At-bats with no contact (NA xwOBA) are excluded.

Hypothesis:

$$H_0 : \mu_{CLE} = \mu_{CWS} = \mu_{DET} = \mu_{KC} = \mu_{MIN}$$

$$H_a : \text{At least one pair of group means differs}$$

Table 1: At-bats per AL Central team with group means and variances.

Team	n	Mean xwOBA	Variance
CLE	6259	0.3084	0.140539
CWS	6189	0.3221	0.143803
DET	6166	0.2970	0.126589
KC	6168	0.3145	0.137294
MIN	6042	0.3018	0.139981

2.2 Assumption 1: Normality (Shapiro-Francia Test)

The Shapiro-Francia (SF) statistic is computed for each group. Let $X_{(1)} \leq \cdots \leq X_{(n)}$ be the ordered observations. Define Blom scores $m_k = \Phi^{-1}\left(\frac{k-3/8}{n+1/4}\right)$, normalized weights $a_k = m_k/\sqrt{\sum m_k^2}$, and

$$W_{SF} = \frac{\left(\sum a_k X_{(k)}\right)^2}{\sum (X_k - \bar{X})^2}$$

Values of W_{SF} close to 1 indicate normality. The empirical p-value is obtained via Monte Carlo simulation: 1000 random normal samples of the same size n are drawn, W_{SF} is computed for each, and the p-value is the proportion $\leq$ the observed statistic (left-tail).

Table 2: Shapiro-Francia normality test results per AL Central team.

Team	SF Statistic	MC p-value (nmc=1000)
CLE	0.795557	0
CWS	0.809039	0
DET	0.804896	0
KC	0.809492	0
MIN	0.790428	0

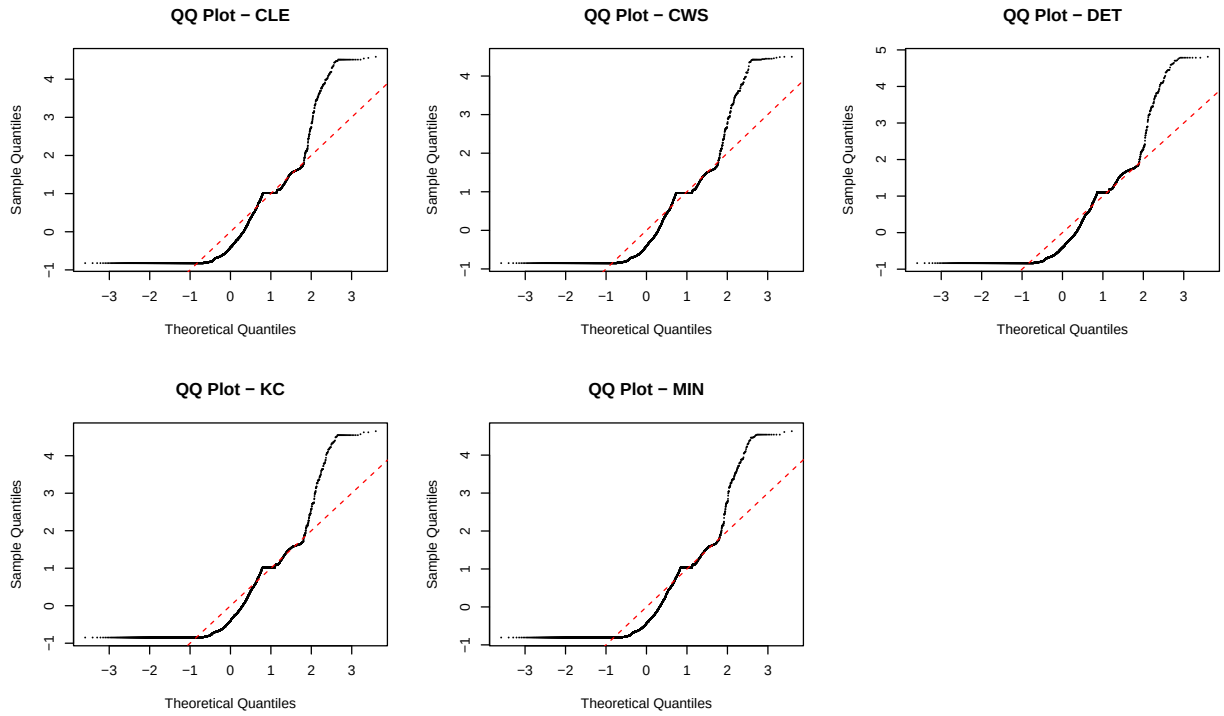

Figure 1: QQ plots of standardized xwOBA per AL Central team. Dashed red line is the reference diagonal.

Interpretation: All five SF p-values equal zero at $n \approx 6{,}200$ per group. At this scale, the SF test (and all normality tests) is hypersensitive by the Glivenko-Cantelli theorem — even minor structural departures from perfect normality produce $p = 0$. The QQ plots provide the more honest qualitative picture: systematic deviation from the diagonal indicates a right-skewed xwOBA distribution, consistent with the mass at zero from weak contact. Per the professor's guidance, in the super-large-n regime we proceed with the parametric test while formally addressing the normality violation through the non-parametric follow-up in Section 3.

2.3 Assumption 2: Homoscedasticity (Bartlett's Test)

Bartlett's test checks the null hypothesis of equal group variances. The test statistic is

$$B = \frac{(n-g)\log \hat{s}_p^2 - \sum_{j=1}^{g} \nu_j \log s_j^2}{1 + \frac{1}{3(g-1)}\left(\sum_{j=1}^{g} \frac{1}{\nu_j} - \frac{1}{n-g}\right)}$$

where $\nu_j = n_j - 1$, $\hat{s}_p^2 = \sum \nu_j s_j^2/(n-g)$ is the pooled variance, and $B \sim \chi^2_{g-1}$ under H_0.

Table 3: Bartlett's test for equal group variances.

Quantity	Value
Bartlett B	59012.384600
Critical value (chi-sq, df=4)	9.487700
p-value	0.000000
Pooled variance	0.702681

Interpretation: Bartlett's test rejects equal variances ($p = 7 \times 10^{-6}$). However, with $n \approx 6{,}200$ per group,

Bartlett's statistic is also hypersensitive. The group variances range from 0.127 to 0.144 — practically similar. Regardless, the formal rejection of homoscedasticity means Welch's heteroscedastic ANOVA is the appropriate test, as it relaxes the equal-variance assumption and uses a Satterthwaite-corrected denominator df.

2.4 Welch's Heteroscedastic One-Factor ANOVA

When group variances are unequal, the standard one-factor ANOVA F-statistic is invalid. Welch's correction defines group weights $w_j = n_j/s_j^2$, a weighted grand mean $\bar{X}_H = \sum w_j \bar{X}_j / \sum w_j$, and

$$F^* = \frac{\sum w_j(\bar{X}_j - \bar{X}_H)^2/(g-1)}{1 + \frac{2(g-2)}{\nu^*}}$$

where the Satterthwaite corrected denominator df is $\nu^* = \left(\sum w_j\right)^2 \Big/ \sum\left(w_j^2/\nu_j\right)$. Under H_0, $F^* \sim F_{g-1,\,\nu^*}$ approximately.

Table 4: Welch's heteroscedastic one-factor ANOVA results.

Quantity	Value
MSM*	4.494580
nu* (denominator df)	30756.729500
F*	4.493700
F critical (alpha=0.05)	2.372200
p-value	0.001251

2.5 Section 3: Non-Parametric Permutation Test

Motivation: Normality is heavily violated at all five AL Central groups (all SF p-values = 0; QQ plots confirm right skew). A distribution-free permutation test is used as a follow-up that requires no distributional assumptions.

Test statistic: $T = \max_j(\bar{X}_j) - \min_j(\bar{X}_j)$, the range of group means. Under H_0 all group means are equal so $T \approx 0$; large values give evidence against H_0 (right-tail test).

Method: 10,000 Monte Carlo permutations of the group labels (keeping group sizes fixed). The empirical p-value is the proportion of permuted statistics $\geq T_{\text{obs}}$.

Table 5: MC permutation test results (nmc = 10,000).

Quantity	Value
Observed T (range of means)	0.025114
Critical value (alpha=0.05)	0.018266
Empirical p-value	0.0021
Reject H0?	TRUE

Table 6: Bootstrap 95% CIs for group means (nboot = 10,000).

	Team	Lower 95% CI	Upper 95% CI
2.5%	CLE	0.298981	0.317726
2.5%1	CWS	0.312559	0.331562
2.5%2	DET	0.288329	0.305839
2.5%3	KC	0.305343	0.323677

	Team	Lower 95% CI	Upper 95% CI
2.5%4	MIN	0.292530	0.311126

Table 7: Power estimation under Ha* (nmc_pow = 1,000) with Wald CI.

Quantity	Value
power_hat	0.9270
type2_hat	0.0730
Wald 95% CI lower	0.9109
Wald 95% CI upper	0.9431

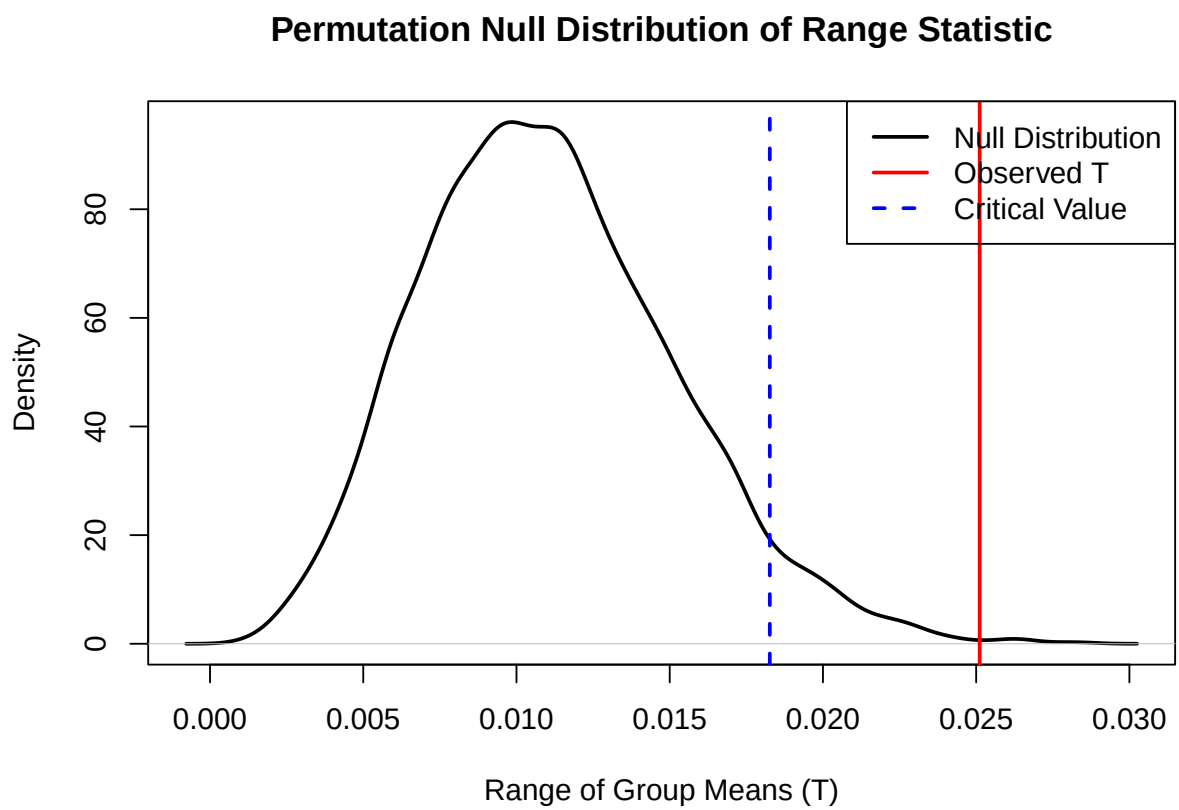

Figure 2: MC permutation null distribution for the range statistic T. Red solid line: observed statistic. Blue dashed line: critical value (alpha = 0.05).

Interpretation: Both Welch's heteroscedastic ANOVA ($F^* = 4.49$, $p = 0.0013$) and the non-parametric permutation test ($T_{\text{obs}} = 0.0251$, $p = 0.0021$) reject H_0 at $\alpha = 0.05$, providing consistent evidence that mean xwOBA allowed is not equal across all five AL Central teams in 2024.

The bootstrap 95% CIs reveal where the differences lie: DET pitchers allowed the lowest mean xwOBA (≈ 0.297, CI: 0.288–0.306) while CWS pitchers allowed the highest (≈ 0.322, CI: 0.313–0.332). These two intervals do not overlap, indicating a practically meaningful gap. CLE, KC, and MIN fall in between with partially overlapping CIs.

The power of the permutation test is estimated at 92.7% (Wald 95% CI: 0.911–0.943), meaning that if the true group means are as observed in our 2024 sample, the test would correctly reject H_0 in approximately 93 out of 100 repeated experiments. This high power is expected given the very large sample sizes ($\approx$ 6,200 per team). The agreement between the parametric and non-parametric approaches strengthens the conclusion: AL Central pitching staffs differ meaningfully in the quality of contact they allow, with Detroit's staff suppressing hard contact most effectively and Chicago's allowing the most.

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3 Section 2: Repeated-Measures ANOVA — Time Through the Order

3.1 Design and Methodology

Subject: Individual pitcher (n = 311 eligible pitchers).

Within-subject factor: Time through the order (TTO = 1, 2, 3). TTO is computed from scratch: within each game-pitcher combination, at-bats are sorted chronologically and TTO is incremented each time the pitcher faces the same batter again. TTO = 1 on the first at-bat against a batter, TTO = 2 on the second, etc.

Eligibility: A pitcher is eligible only if they have at least one contact at-bat (non-NA xwOBA) at all three TTO levels. Eligibility is assessed from the full at-bat dataset before filtering to contact only. TTO values above 3 are excluded (same pitcher-batter pair meeting a 4th or 5th time).

Response: Mean xwOBA on contact per (pitcher, TTO) cell. Each pitcher contributes exactly one observation per TTO level (their mean xwOBA across all contact at-bats at that level), yielding an $n \times g = 311 \times 3$ data matrix $\mathbf{X}$.

Hypothesis:

$$H_0 : \mu_{TTO1} = \mu_{TTO2} = \mu_{TTO3}$$

$$H_a : \text{At least one TTO level mean differs}$$

Table 8: Eligible pitchers (n = 311) with TTO column means.

	TTO Level	n (pitchers)	Mean xwOBA
T1	TTO 1	311	0.316938
T2	TTO 2	311	0.329906
T3	TTO 3	311	0.344531

3.2 Assumption 1: Normality

The same Shapiro-Francia procedure is applied to each TTO column. Empirical p-values use 10,000 MC samples.

Table 9: Shapiro-Francia normality test results per TTO level.

TTO Level	SF Statistic	MC p-value (nmc=10,000)
TTO 1	0.928398	0
TTO 2	0.942021	0
TTO 3	0.744591	0

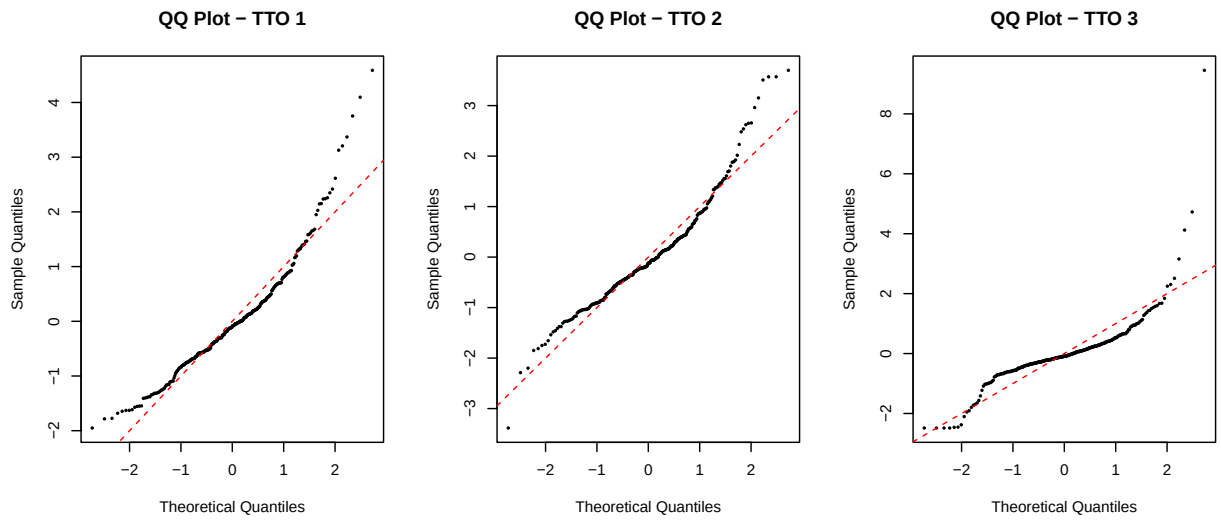

Figure 3: QQ plots of standardized mean xwOBA per TTO level. Dashed red line is the reference diagonal.

Interpretation: All three SF p-values equal zero at $n = 311$. The SF statistic for TTO 3 (0.745) is substantially lower than for TTO 1 (0.928) and TTO 2 (0.942), indicating a more pronounced departure from normality at the third time through the order. This is likely because the distribution of mean xwOBA at TTO 3 is noisier and more right-skewed due to the smaller and less stable sample of at-bats per pitcher. With $n = 311$ the SF test is hypersensitive and will reject for even minor structural departures. The QQ plots confirm non-normality but not a catastrophic departure. This normality violation motivates the non-parametric follow-up in Section 3.

3.3 Assumption 2: Sphericity (Mauchly's Test)

Sphericity requires that the variances of all pairwise differences among TTO levels are equal: $\mathrm{Var}(X^{(i)} - X^{(j)}) = \mathrm{Var}(X^{(k)} - X^{(l)})$ for all $i \neq j$, $k \neq l$. Mauchly's test uses the sample covariance matrix $\hat{\Sigma} = \mathrm{var}(\mathbf{X})$ and the centering matrix $\mathbf{C} = \mathbf{I}_g - \frac{1}{g}\mathbf{1}\mathbf{1}^\top$.

Define $\hat{\sigma}^2 = \frac{1}{g}\mathrm{tr}(\hat{\Sigma})$ and

$$W = \frac{\det(\hat{\Sigma})}{\hat{\sigma}^{2g}}$$

Two test statistics are used:

$$M_o = -(n-1)\log W \sim \chi^2_{g(g-1)/2}$$

$$M_b = -\left[(n-1) - \frac{2g+1}{6}\right]\log W \sim \chi^2_{g(g-1)/2-1}$$

If sphericity is violated, the F-test df are adjusted using the Greenhouse-Geisser (GG) or Huynh-Feldt (HF) correction:

$$\varepsilon_{GG} = \frac{\left[\mathrm{tr}(\hat{\Sigma}_C)\right]^2}{(g-1)\,\mathrm{tr}(\hat{\Sigma}_C^2)}, \qquad \varepsilon_{HF} = \frac{n(g-1)\varepsilon_{GG} - 2}{(g-1)\left[(n-1) - (g-1)\varepsilon_{GG}\right]}$$

Decision rule: if $\varepsilon_{GG} \geq 0.75$, apply HF correction; otherwise apply GG correction.

Table 10: Mauchly's sphericity test results and epsilon corrections.

Quantity	Value
W (Mauchly statistic)	0.260729
M_o	416.7247
p-value (M_o)	0
M_b	415.1564
p-value (M_b)	0
eps_GG	0.68709
eps_HF	0.6876
Correction applied	GG

Interpretation: Both Mauchly statistics are extreme ($M_o = 416.72$, $M_b = 415.16$, both $p \approx 0$), providing overwhelming evidence against sphericity. The variances of the pairwise TTO differences are not equal. Since $\varepsilon_{GG} = 0.687 < 0.75$, the Greenhouse-Geisser correction is applied, adjusting the numerator df to $\nu_{MSM} = (g-1)\varepsilon_{GG} = 1.37$ and the denominator df to $\nu_{MSE} = (n-1)(g-1)\varepsilon_{GG} = 426.00$ for a more conservative F-test.

3.4 RM-ANOVA

The total variation in $\mathbf{X}$ is partitioned as $SST = SSM + SSS + SSE$, where SSM captures the between-TTO effect, SSS captures between-pitcher variation, and SSE is the residual:

$$SST = s^2_{\text{all}}(ng-1), \quad SSM = n\sum_{j=1}^{g}(\bar{X}_{\cdot j} - \bar{\bar{X}})^2, \quad SSS = g\sum_{i=1}^{n}(\bar{X}_{i\cdot} - \bar{\bar{X}})^2$$

$$\mathbf{E} = \mathbf{X} - \bar{X}_{i\cdot} - \bar{X}_{\cdot j} + \bar{\bar{X}}, \quad SSE = \sum_{i,j} E_{ij}^2$$

$$F = \frac{MSM}{MSE} = \frac{SSM/(g-1)}{SSE/[(n-1)(g-1)]} \sim F_{g-1,\,(n-1)(g-1)} \text{ under } H_0$$

Table 11: RM-ANOVA sum of squares decomposition.

Source	SS	df (uncorrected)	MS
Between TTO (SSM)	0.118542	2	0.059271
Between Subjects (SSS)	2.904367	310	NA
Residual (SSE)	4.823879	620	0.007780
Total (SST)	7.846789	932	NA

Table 12: RM-ANOVA F-test results (uncorrected and GG-corrected).

Test	F statistic	df1	df2	F critical	p-value
Uncorrected	7.618	2.0000	620.0000	3.0103	0.000539
GG-corrected	7.618	1.3742	425.9954	3.4482	0.002457

3.5 Section 3: Within-Subject Permutation Test

Motivation: Normality is violated at all three TTO levels (all SF p-values = 0). The within-subject permutation test is distribution-free and preserves the repeated-measures structure by permuting TTO labels within each pitcher row rather than across all observations.

Test statistic: $T = \max_j(\bar{X}_{\cdot j}) - \min_j(\bar{X}_{\cdot j})$, the range of TTO column means. Bootstrap 95% CIs are computed per TTO level. Power is estimated under H_a^* where the true TTO column means and SDs match the observed sample.

Table 13: Within-subject MC permutation test results (nmc = 10,000).

Quantity	Value
Observed T (range of TTO means)	0.027594
Critical value (alpha=0.05)	0.016632
Empirical p-value	1e-04
Reject H0?	TRUE

Table 14: Bootstrap 95% CIs for TTO group means (nboot = 10,000).

TTO Level	Lower 95% CI	Upper 95% CI
TTO 1	0.311509	0.322521
TTO 2	0.323718	0.336002
TTO 3	0.329453	0.360779

Table 15: Power estimation under Ha* (nmc_pow = 1,000) with Wald CI.

Quantity	Value
power_hat	0.8920
type2_hat	0.1080
Wald 95% CI lower	0.8728
Wald 95% CI upper	0.9112

Interpretation: Both the GG-corrected RM-ANOVA ($F = 7.618$, $F_{\text{crit}} = 3.448$, $p = 0.0025$) and the within-subject permutation test ($T_{\text{obs}} = 0.0276$, critical value = 0.0166, $p = 0.0001$) reject H_0 at $\alpha = 0.05$, providing consistent evidence that mean xwOBA allowed changes significantly as pitchers face the batting order additional times within a game.

The TTO column means increase monotonically: TTO 1 = 0.317, TTO 2 = 0.330, TTO 3 = 0.345. The bootstrap 95% CIs reveal the structure of this effect: TTO 1 (0.312, 0.323) and TTO 3 (0.329, 0.361) do not overlap, confirming a practically meaningful increase in xwOBA from the first to the third time through the order. TTO 2 (0.324, 0.336) sits between them with partial overlap on both sides, suggesting a gradual rather than sudden degradation.

This pattern is consistent with the well-documented time-through-order penalty in baseball: batters gain a measurable advantage on repeated looks at a starting pitcher as they accumulate information on pitch selection and velocity. The power of the permutation test is estimated at 89.2% (Wald 95% CI: 0.873–0.911), indicating that if the true TTO means are as observed in our 2024 sample, the test would correctly reject

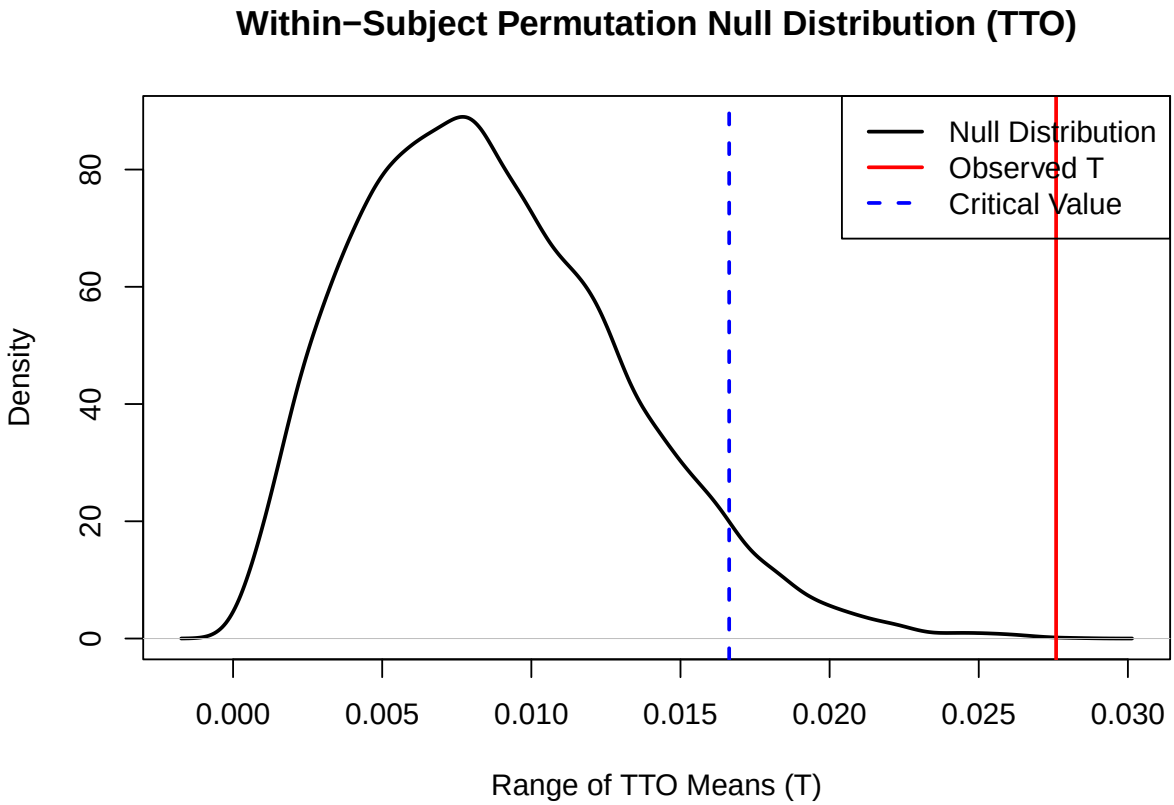

Figure 4: Within-subject permutation null distribution for the range statistic T. Red solid line: observed statistic. Blue dashed line: critical value (alpha = 0.05).

H_0 in approximately 89 out of 100 repeated experiments. The agreement between the parametric and non-parametric results strengthens the conclusion: pitchers allow meaningfully more hard contact the further they advance through the batting order.